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HT Board Booster

Class X Science

Physics

CBSE Sample Paper Question Bank

Chapter wise Practice

Chapters 3 - 4
Electricity and Magnetic

Session 2026-27

Practice | Perform | Progress

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Hunny Gupta

Class X Physics
CBSE SQP 2019-2025
Ch- Electricity

SECTION –A

Q.1. When a 4V battery is connected across an unknown resistor there is a current of 100 mA in the circuit. The value of the resistance of the resister is: **(CBSE SQP 2019-20)**

(a) 4 Ω (b) 40 Ω (c) 400 Ω (d) 0.4 Ω

Q.2. Unit of electric power may also be expressed as: **(CBSE SQP 2019-20)**

(a) volt-ampere (b) kilowatt-hour
(c) watt-second (d) joule-second

Q.3. Two statements are given- one labeled *Assertion* (A) and the other labeled *Reason* (R). Select the correct answer to these questions from the codes (i), (ii), (iii) and (iv) as given below

i) Both A and R are true and R is correct explanation of the assertion.
ii) Both A and R are true but R is not the correct explanation of the assertion.
iii) A is true but R is false.
iv) A is false but R is true.

Assertion: A fuse wire is always connected in parallel with the mainline. **(CBSE SQP 2019-20)**
Reason: If a current larger than the specified value flows through the circuit, fuse wire melts.

Q.4. Two unequal resistances are connected in parallel. If you are not provided with any other parameters (eg. numerical values of I and R), what can be said about the voltage drop across the two resistors? **(CBSE SQP 2020-21)**

OR

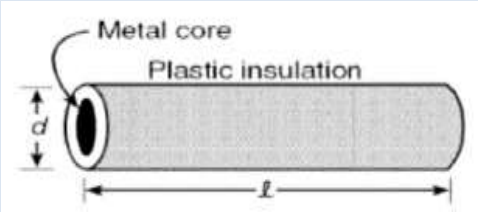
Some work is done to move a charge Q from infinity to a point A in space. The potential of the point A is given as V. What is the work done to move this charge from infinity in terms of Q and V?

Q.5. A complete circuit is left on for several minutes, causing the connecting copper wire to become hot. As the temperature of the wire increases, the electrical resistance of the wire.

(a) decreases. (b) remains the same. **(CBSE SQP 2022-23)**
(c) increases. (d) increases for some time and then decreases.

Q.6. Plastic insulation surrounds a wire having diameter d and length I as shown above. A decrease in the resistance of the wire would be produced by an increase in the **(CBSE SQP 2022-23)**

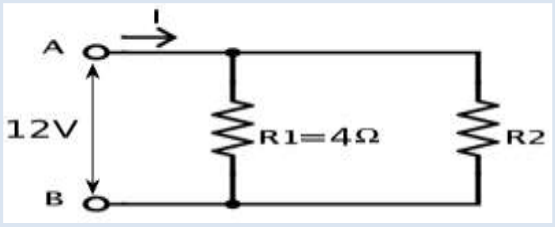
(a) length 1 of the wire
(b) diameter d of the wire
(c) temperature of the wire
(d) thickness of the plastic insulation



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SECTION –B

Q.7. A student has two resistors- 2 Ω and 3 Ω. She has to put one of them in place of R2 as shown in the circuit. The current that she needs in the entire circuit is exactly 9A. Show by calculation which of the two resistors she should choose.

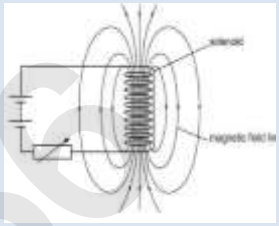


(CBSE SQP 2020-21)

Q.8. A piece of wire of resistance R is cut into three equal parts. These parts are then connected in parallel. If the equivalent resistance of this parallel combination is R_1, what is the value of the ratio R_1: R?

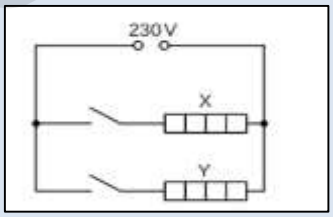
OR

Refer to the image below and state how the magnetic field pattern indicates regions where the magnetic field is stronger outside the magnet? What happens to the magnetic field when the current in the circuit is reversed? **(CBSE SQP 2023-24)**



Q.9. **Attempt either option A or B.** **(CBSE SQP 2024-25)**

(a) The electric circuit (above figure) in a clothes dryer contains two heaters X and Y in parallel. The above figure shows the circuit connected to a 230 V power supply. When both switches are closed, the current in X is 3.5 A.



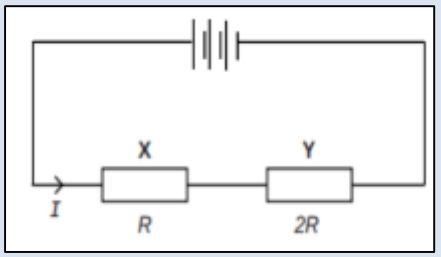
Analyse the circuit given above and answer the following questions.

(i) Calculate the power developed in heater X.

(ii) If the resistance of X is double that of Y calculate the current in heater Y.

OR

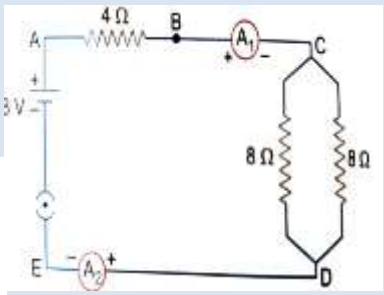
(b) The above figure shows two resistors X and Y connected in series to a battery. The power dissipated for this combination is P1. When these resistors are connected in parallel to the same battery then the power dissipated is given by P2. Find out the ratio $\frac{P_1}{P_2}$.



Q.10. Find out the following in the electric circuit given in the figure-

(i) Effective resistance of two 8 ohm resistors in the combination.

(ii) Current flowing through the 4-ohm resistor.

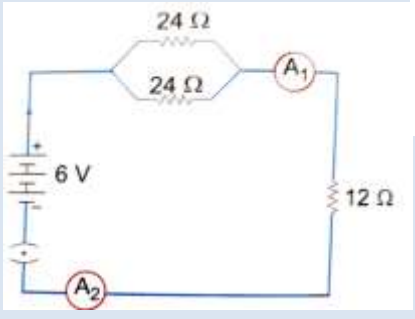


OR

Study the circuit and find out-

(i) Current in 12 Ω resistor

(ii) Difference in the readings of ammeter A_1 and A_2 if any.



SECTION –C

Q.11. Suppose your parents have constructed a two room house and you want that in the living room there should be a provision of one electric bulb, one electric fan, a refrigerator and a plug point for appliances of power up to 2 kilowatts. Draw a circuit diagram showing electric fuse and earthing as safety devices.

(CBSE SQP 2019-20)

Q.12. (i) State the law that explains the heating effect of current with respect to the measurable properties in an electrical circuit. **(CBSE SQP 2023-24)**

(ii) List the factors on which the resistance of a conductor depends.

Q.13. (a) State the relationship between the resistance R of a wire to its length I and cross sectional area A. Use the mathematical symbols to arrive at the final formula.

(b) Using the formula to define the resistivity of a material. **(CBSE SQP 2024-25)**

Q.14. A student needs to make a 0.12 Q resistor. She has some copper wire of 0.80 mm diameter. Resistivity of copper is 1.8×10^{-8} Ω m. **(CBSE SQP 2025-26)**

(i) Determine the cross-sectional area of the wire.

(ii) Calculate the length of wire required for the 0.12 Ω resistor

Section –D

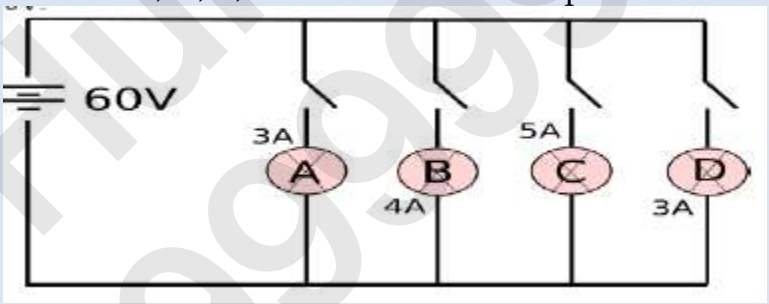
Q.15. (i) Consider a conductor of resistance 'R', length 'L', thickness'd' and resistivity 'ρ'. Now this conductor is cut into four equal parts. What will be the new resistivity of each of these parts? Why? **(CBSE SQP 2019-20)**

(ii) Find the resistance if all of these parts are connected in:

(a) Parallel (b) Series

(iii) Out of the combinations of resistors mentioned above in the previous part, for a given voltage which combination will consume more power and why?

Q.16. In the given circuit, A, B, C and D are four lamps connected with a battery of 60V.



Analyse the circuit to answer the following questions. **(CBSE SQP 2020-21)**

(i) What kind of combination are the lamps arranged in (series or parallel)?

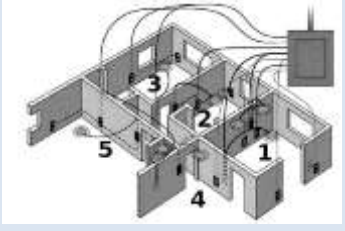
(ii) Explain with reference to your above answer, what are the advantages (any two) of this combination of lamps?

(iii) Explain with proper calculations which lamp glows the brightest?

(i) Find out the total resistance of the circuit.

Q.17. The diagram above is a schematic diagram of a household circuit. The house shown in the above diagram has 5 usable spaces where electrical connections are made. For this house, the mains have a voltage of 220 V and the net current coming from the mains is 22A.

(CBSE SQP 2022-23)



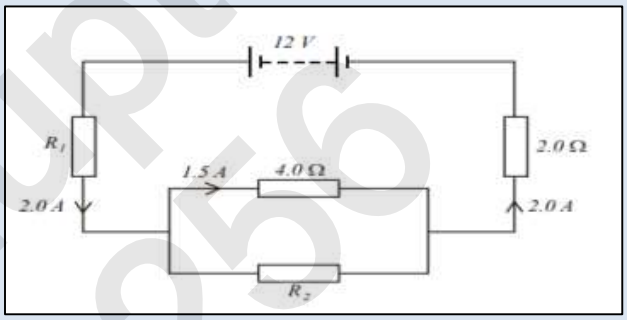
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(a) What is the mode of connection to all the spaces in the house from the mains?
(b) The spaces 5 and 4 have the same resistance and spaces 3 and 2 have respective resistances of 20 Ω and 30 Ω. Space 1 has a resistance double that of space 5. What is the net resistance for space 5?
(c) What is the current in space 3?
(d) What should be placed between the main connection and the rest of the house's electrical appliances to save them from accidental high electric current?

Q.18. **Attempt either option A or B**

The above circuit is a part of an electrical device. Use the information given in the question to calculate the following. **(CBSE SQP 2024-25)**

(i) Potential Difference across R2.
(ii) Value of the resistance R2.
(iii) Value of resistance R1

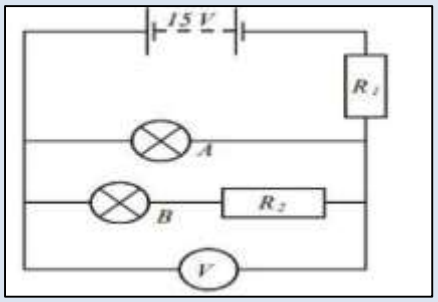


OR

As shown in the figure above A and B are two lamps. Lamp A is rated at 12 V, 24W. Lamp B is rated at 6.0 V. When lamp B operates at its rated voltage, the current in it is 3.0 A. The values of R1 and R2 are chosen so that both lamps operate at their rated voltages.

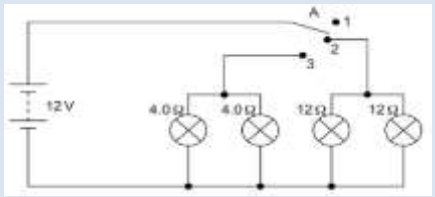
Based on the information given, answer the following.

(i) Calculate the current in Lamp A.
(ii) State and give reason for the reading of the Voltmeter.
(iii) Calculate the resistance of R2.
(iv) Find the value of the resistance R1.



SECTION –E

Q.19. Vinita and Ahmed demonstrated a circuit that operates the two headlights and the two sidelights of a car, in their school exhibition. Based on their demonstrated circuit, answer the following questions.



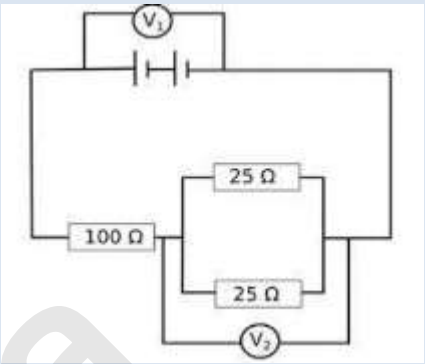
(i) State what happens when switch A is connected to
(a) Position 2
(b) Position 3
(ii) Find the potential difference across each lamp when lit.
(iii) Calculate the current
(a) in each 12 Ω lamp when lit.
(b) In each 4 Ω lamp when lit. **(CBSE SQP 2023-24)**

OR

(iv) Show, with calculations, which type of lamp, 4.0 Ω or 12 Ω, has the higher power.

Q.20. The arrangement of resistors shown in the above figure is connected to a battery.;
The power dissipation in the 100 Ω resistor is 81 W.
Calculate

(i) the current in the circuit
(ii) the reading in the voltmeter V_2.
(iii) the reading in the voltmeter V_1.

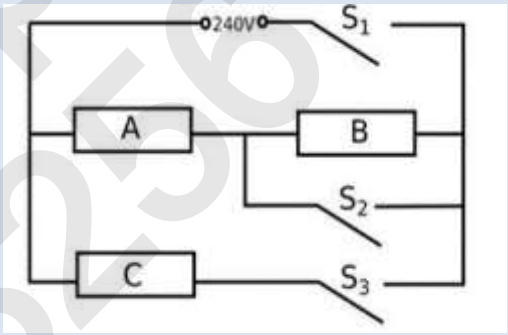


OR

An electric heater consists of three similar heating elements A, B and C, connected as shown in the figure above. Each heating element is rated as 1.2kW, 240 V and has constant resistance. S1, S2 and S3 are respective switches.

The circuit is connected to a 240 V supply.

(i) Calculate the resistance of one heating element.
(ii) Calculate the current in each resistor when only S1 and S3 are closed.
(iii) Calculate the power dissipated across A when S1, S2 and S3 are closed **(CBSE SQP 2025-26)**

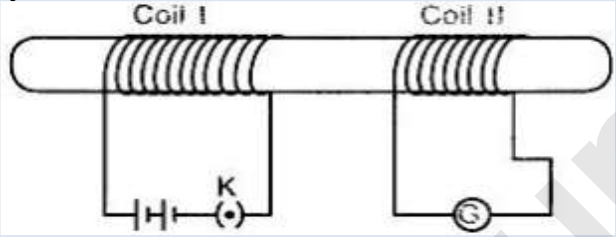


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Class X Physics
CBSE SQP 2019-2025
Ch- Magnetic Effect of Electric Current

Section A

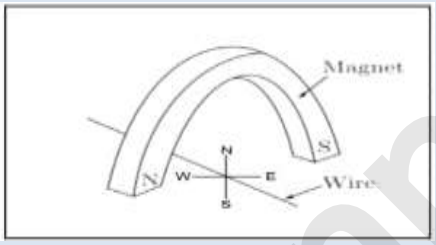
Q.1. In the arrangement shown in figure there are two coils wound on a non-conducting cylindrical rod. Initially the key is not inserted in the circuit. Later the key is inserted and then removed shortly after. **(CBSE SQP 2020-21)**



What are the two observations that can be noted from the galvanometer reading?

Q.2. Draw the magnetic field lines around a straight current carrying conductor. **(CBSE SQP 2020-21)**

Q.3. A copper wire is held between the poles of a magnet. **(CBSE SQP 2022-23)**



The current in the wire can be reversed. The pole of the magnet can also be changed. over. In how many of the four directions shown can the force act on the wire?

(a) 1 (b) 2 (c) 3 (d) 4

Q.4. Which of the following pattern correctly describes the magnetic field around a long straight wire carrying current? **(CBSE SQP 2022-23)**

(a) straight lines perpendicular to the wire.
(b) straight lines parallel to the wire.
(c) radial lines originating from the wire.
(d) concentric circles centred around the wire.

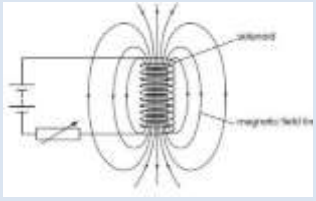
Q.5. **Assertion (A):** On freely suspending a current carrying solenoid, it comes to rest in Geographical N-S direction. **(CBSE SQP 2022-23)**
Reason(R) : One end of current carrying straight solenoid behaves as a North pole and the other end as a South pole, just like a bar magnet.

Q.6. **Assertion (A):** A compass needle is placed near a current carrying wire. The deflection of the compass needle decreases when the magnitude of the current in the wire is increased.
Reason(R): The strength of a magnetic field at a point near the conductor increases on increasing the current. **(CBSE SQP 2023-24)**

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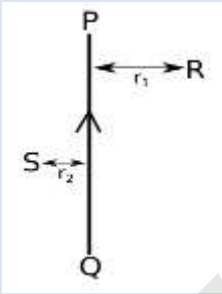
Section – B

Q.7. Refer to the image below and state how the magnetic field pattern indicates regions where the magnetic field is stronger outside the magnet? What happens to the magnetic field when the current in the circuit is reversed? **(CBSE SQP 2023-24)**



Section – C

Q.8. PQ is a current carrying conductor in the plane of the paper as shown in the figure below.



(i) Find the directions of the magnetic fields produced by it at points R and S?
(ii) Given $r_1 > r_2$, where will the strength of the magnetic field be larger? Give reasons.
(iii) If the polarity of the battery connected to the wire is reversed, how would the direction of the magnetic field be changed?
(iv) Explain the rule that is used to find the direction of the magnetic field for a straight current carrying conductor.

Q.9. A student fixes a white sheet of paper on a drawing board. He places a bar magnet in the centre and sprinkles some iron filings uniformly around the bar magnet. Then he taps gently and observes that iron filings arrange themselves in a certain pattern. **(CBSE SQP 2022-23)**

(a) Why do iron filings arrange themselves in a particular pattern?
(b) Which physical quantity is indicated by the pattern of field lines around the bar magnet?
(c) State any two properties of magnetic field lines.

OR

A compass needle is placed near a current carrying wire. State your observations for the following cases and give reasons for the same in each case-

(a) Magnitude of electric current in wire is increased.
(b) The compass needle is displaced away from the wire.

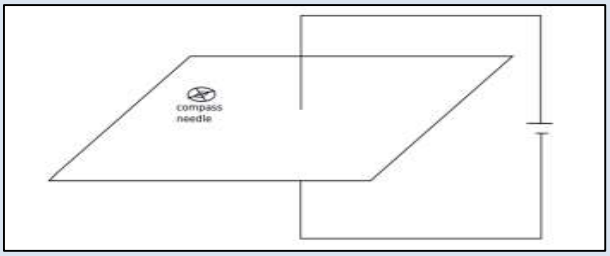
Q.10. Anannya responded to the question: Why do electrical appliances with metallic bodies are connected to the mains through a three pin plug, whereas an electric bulb can be connected with a two pin plug? **(CBSE SQP 2022-23)**

She wrote: Three pin connections reduce heating of connecting wires.

(i) Is her answer correct or incorrect? Justify.
(ii) What is the function of a fuse in a domestic circuit?

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Q.11. (a)



Mona was doing an experiment with a magnetic compass and a straight current-carrying wire. She observed that as she moved the compass away from the current-carrying wire, the deflection of the compass needle reduced. **(CBSE SQP 2024-25)**

(a) Explain why the deflection of the compass needle reduced as Mona moved away the compass needle from the current carrying wire.

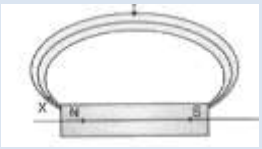
(b) Mention one thing that could have changed in the circuit of the wire that could increase the deflection of the needle.

(c) Explain with reason what will be the direction of the magnetic field associated with the wire for the case described by the above figure.

Q.12. Magnetic field lines are shown in the given diagram. A student makes a statement that the magnetic field at X is stronger than at Y. **(CBSE SQP 2025-26)**

(i) Explain with reason if the student's claim is correct.

(ii) Also redraw the diagram and mark the direction of magnetic field lines.



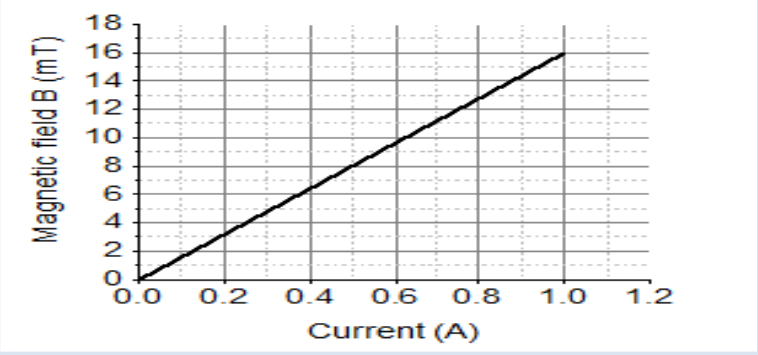
Section – E

Q.13. Read the following and answer any 4 questions from 20 (i) to 20 (v). **(CBSE SQP 2020-21)**

A solenoid is a long helical coil of wire through which a current is run in order to create a magnetic field. The magnetic field of the solenoid is the superposition of the fields due to the current through each coil. It is nearly uniform inside the solenoid and close to zero outside and is similar to the field of a bar magnet having a north pole at one end and a south pole at the other depending upon the direction of current flow. The magnetic field produced in the solenoid is dependent on a few factors such as, the current in the coil, number of turns per unit length etc.

The following graph is obtained by a researcher while doing an experiment to see the variation of the magnetic field with respect to the current in the solenoid.

The unit of magnetic field as given in the graph attached is in milli-Tesla (mT) and the current is given in Ampere.



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(i) What type of energy conversion is observed in a linear solenoid? (1)

a. Mechanical to Magnetic b. Electrical to Magnetic
c. Electrical to Mechanical d. Magnetic to Mechanical

(ii) What will happen if a soft iron bar is placed inside the solenoid? (1)

a. The bar will be electrocuted resulting in short-circuit.
b. The bar will be magnetised as long as there is current in the circuit.
c. The bar will be magnetised permanently.
d. The bar will not be affected by any means.

(iii) The magnetic field lines produced inside the solenoid are similar to that of … (1)

a. a bar magnet b. a straight current carrying conductor
c. a circular current carrying loop d. electromagnet of any shape

(iv) After analysing the graph a student writes the following statements. (1)

I. The magnetic field produced by the solenoid is inversely proportional to the current.
II. The magnetic field produced by the solenoid is directly proportional to the current.
III. The magnetic field produced by the solenoid is directly proportional to square of the current.
IV. The magnetic field produced by the solenoid is independent of the current.

Choose from the following which of the following would be the correct statement(s).

a. Only IV b. I and III and IV
c. I and II d. Only II

(v) From the graph deduce which of the following statements is correct. (1)

a. For a current of 0.8A the magnetic field is 13 mT
b. For larger currents, the magnetic field increases non-linearly.
c. For a current of 0.8A the magnetic field is 1.3 mT
d. There is not enough information to find the magnetic field corresponding to 0.8A current.

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